The above examples suggest that there are two integral square roots for a perfect square number. However, in working out the problems, we will take up only positive integral square root. The positive square root of a number is always denoted by the symbol $\sqrt{\ }$. Thus, $\sqrt{4}$ is 2 (and not –2). Also, $\sqrt{9}$ is 3 (and not –3). We have to remember that this is a universally accepted notation.

1.7.1 Square root through Prime Factorisation

Study the following table giving the prime factors of numbers and those of their squares.

Numbers	Prime factorisation of the numbers	The square of given numbers	Prime factorisation of their squares
6	$6 = 2 \times 3$	$6^2 = 36$	$36 = 2 \times 2 \times 3 \times 3 = (2 \times 3)^2$
8	$8 = 2 \times 2 \times 2$	$8^2 = 64$	$64 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 = (2 \times 2 \times 2)^2$
12	$12 = 2 \times 2 \times 3$	$12^2 = 144$	$44 = 2 \times 2 \times 2 \times 2 \times 3 \times 3 = (2 \times 2 \times 3)^2$
15	$15 = 3 \times 5$	$15^2 = 225$	$225 = 3 \times 3 \times 5 \times 5 = (3 \times 5)^2$

Look at 6 and its prime factors. How many times do 2 and 3 occur in 36 in the list? Now, look at its square 36 and its prime factors. How many times do 2 and 3 occur in 36 here?

Repeat the above task in the case of other numbers 8, 12, and 15 also. (We may also choose our own numbers and their squares). What do we find? We find that,

The number of times a prime factor occurs in the square of a number = twice the number of times it occurs in the prime factorisation of the number.

We use this idea to find the square root of a square number. First, resolve the given number into prime factors. Group the identical factors in pairs and then take one from them to find the square root.

Example 1.24

Find the square root of 324 by prime factorisation.

Solution:

First, resolve the given number into prime factors. Group the identical factors in pairs and then take one from them to find the square root.

Now, $324 = 2 \times 2 \times 3 \times 3 \times 3 \times 3$

$= 2^2 \times 3^2 \times 3^2$

$= (2 \times 3 \times 3)^2$

$\therefore \sqrt{324} = \sqrt{(2 \times 3 \times 3)^2}$

$= 2 \times 3 \times 3$

$\therefore \sqrt{324} = 18$

2	324
2	162
3	81
3	27
3	9
3	3
	1

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